

Bob Ricardy

📍 Singapore | ✉ bobbyricardy@gmail.com | ☎ +65 9366 1759 | 🌐 bobbyricardy.github.io | 🌐 bobricardy | 🌐 bobbyricardy

Professional Summary

Full-stack engineer with experience in banking, building internal tools end-to-end — backend services, React frontends, and custom Elasticsearch search and visualization apps. Led engineering teams for the last four years, bringing order to fragmented systems by standardizing data from multiple regional markets into a single internal platform.

Skills

Full-Stack Engineering: JavaScript, React, Node.js, Python, Java, REST APIs

Observability & Telemetry: Elasticsearch, Logstash, Ruby (scripting/data pipelines), Kibana, Grafana, Vega, Filebeat

Data & Messaging: Apache Kafka, ETL Pipelines, Data Modeling, SQL, NoSQL

Systems & Operations: Enterprise Linux, Bash/Shell Scripting, CI/CD Pipelines, High Availability (HA), Incident Triage, Git

Modern Tooling & AI: Claude Code, Agentic Coding Workflows

Experience

Vice President (Software Engineering Lead), DBS Bank

Mar 2024 – Apr 2025

- Managed a team of software and platform engineers, overseeing sprint delivery and technical mentoring for direct reports
- Prototyped and evaluated dynamic layout algorithms (React Flow) to automatically visualize multi-tier service dependencies for the team's topology visualization framework
- Led the delivery of an end-to-end service observability and journey-monitoring platform (React, Node.js, Elasticsearch, Kafka) across 130+ critical digital banking customer journeys as part of a bank-wide system resilience initiative
- Partnered with Lines of Business across regional markets to standardize telemetry specifications around core RED/Golden Signals (Response time, Availability, Throughput, Error rates), enabling teams to stream structured data into centralized dashboards
- Built a real-time geospatial monitoring platform (React, Node.js) mapping 70+ retail branches, ingesting telemetry directly from the internal branch operations platform to evaluate site health in <15 minutes

Assistant Vice President (Software Engineering Lead), DBS Bank

Mar 2021 – Feb 2024

- Scaled the team's logging infrastructure across a 21-node Elasticsearch cluster ingesting 100GB+ daily logs, tuning Logstash batch pipelines and indexing buffers to sustain <5-minute rolling dashboard query latency
- Led data engineering for a centralized IT service intelligence platform, designing ETL pipelines to ingest and normalize ticketing telemetry across disparate ITSM sources and shipping the MVP within 2 months
- Engineered real-time monitoring web applications (React, Node.js) mapping 1,000+ nationwide ATMs, correlating mainframe status data with core banking inventory and geocoded coordinates
- Built an automated SLA reporting engine (Node.js, Vega, Kibana) delivering scheduled metric breakdowns to leadership, identifying cross-departmental incident resolution bottlenecks
- Developed batch ingestion workflows delivering 6+ enterprise telemetry datasets into the bank's Hadoop infrastructure and built an incident calculation tool to evaluate potential outage severity
- Executed zero-downtime Elasticsearch and Grafana upgrades across multiple major versions, handling schema transitions without interrupting data ingestion

Senior Associate / Associate (Software Engineer), DBS Bank

May 2018 – Feb 2021

- Engineered and scaled a distributed ETL and event streaming pipeline across a 7-node cluster (3x Kafka brokers, 3x Elasticsearch data nodes, Logstash), processing an estimated ~1GB+ (~1M+ events) of daily operational telemetry
- Re-engineered an open-source network topology engine (Netflix-Vizceral, React, Node.js) to map live inter-market traffic across 6 core markets (SG, HK, TW, CN, IN, ID)
- Authored custom Logstash Ruby filter plugins and Grok regex patterns to parse and normalize legacy banking logs extracted via Filebeat across 100+ servers, enforcing strict data schemas and establishing deadman alerts on zero-volume indices
- Patched internal Grafana server code for enterprise TLS cipher suite compliance, and added configurable 0/null value differentiation to single-stat panels, preventing real outages from displaying as a false "healthy" status
- Developed internal diagnostic full-stack web applications (React, Node.js) and 10+ real-time operational dashboards (Grafana, Kibana), owning Linux host environments and transitioning release workflows from manual scripts to automated CI/CD pipelines

Application Consultant (Client: DBS Bank), NCS Group

Aug 2016 – May 2018

- Built early Elasticsearch (v2.0)-based monitoring and search tooling before native visualization tools had matured
- Deployed and managed early cluster infrastructure and Filebeat shipper agents for application log harvesting

Projects

Rotom-Dex — Personal Observability Platform – bobbyricardy.github.io

2026 – present

End-to-end observability and APM telemetry platform engineered for a personal portfolio, mirroring production-grade distributed architectures.

- Built an edge telemetry pipeline using Cloudflare Workers and secure Tunnels to stream Real User Monitoring (RUM) metrics and APM traces to Elasticsearch.
- Authored Ansible playbooks with Ansible Vault to automate Debian host provisioning, service orchestration, and secure secret handling.
- Migrated host infrastructure from GCP to DigitalOcean, cutting cloud hosting costs by over 50% with zero data loss
- Engineered a public REST API documented with Swagger alongside a real-time 30-day uptime heatmap

Education

BEng **Nanyang Technological University**, Computer ScienceSingapore
Aug 2013 – Jun 2016**Diploma** **Temasek Polytechnic**, Electronics (Robotics) – with MeritSingapore
Apr 2007 – Sep 2010